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## SYSTEM FOR REMOVABLY FASTENING A SCREEN TO AN INSTRUMENT PANEL OF AN AIRCRAFT, AND VISUALIZATION SYSTEM COMPRISING SUCH A FASTENING SYSTEM


### Abstract

A fastening system comprises a rail, a fastening device to fasten the rail to an instrument panel of a cockpit of an aircraft without modification of the instrument panel, and at least one movable carriage provided with a support device and configured to move in translation along the rail and to be brought into an operation position, the support device being provided with a support bracket to bear the screen and to be raised into the operation position and lowered on the movable carriage during a translational movement of the movable carriage on the rail.



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## Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS
[0001] This application claims the benefit of French Patent Application Number 2402009 filed on Feb. 29, 2024, the entire disclosure of which is incorporated herein by way of reference.
FIELD OF THE INVENTION
[0002] The present invention relates to a system for removably fastening a screen to an instrument panel of a cockpit of an aircraft, and to a visualization system comprising a screen and such a fastening system.
BACKGROUND OF THE INVENTION
[0003] It is known that it may be useful or necessary to install an additional screen on the instrument panel of an aircraft, in particular of a transport plane.
[0004] More particularly, although not exclusively, it may be a screen intended to help with carrying out tests during test flights of the aircraft. Such a test screen may, for example, be used in an on-board simulation system, such as described in patent application FR2005575. This test screen is disposed in front of a customary screen of the aircraft during a given phase of the test flight and can be retracted when it is no longer being used in order to free up the customary screen.
[0005] Since the use of an additional screen is in principle temporary, there is a need to find a solution which makes it possible to install an additional screen on the instrument panel of a cockpit of an aircraft, with simple and rapid installation and also simple and rapid deinstallation, without having to modify the equipment of the instrument panel.
SUMMARY OF THE INVENTION
[0006] It is an object of the present invention to propose a solution for meeting the aforementioned need. The present invention relates to a system for removably fastening at least one screen to an instrument panel of a cockpit of an aircraft.
[0007] According to the invention, said fastening system comprises: [0008] a rectilinear rail having a longitudinal axis; [0009] a fastening device intended to fasten the rail to the instrument panel without modification of said instrument panel; and [0010] at least one movable carriage provided with a support device, said movable carriage being configured to be able to be moved in translation along said rail and to be brought at least into what is referred to as an operation position, said support device being provided with a support bracket intended to bear the screen, the support bracket being able to pivot about an axis parallel to said longitudinal axis and being configured to, by the pivoting thereof, be able to be raised into what is referred to as a raised configuration, at least in said operation position, and to be able to be lowered on the movable carriage into what is referred to as a lowered configuration, at least during a translational movement of said movable carriage on the rail.
[0011] Thus, by virtue of the invention, the fastening system and also an additional screen borne by this fastening system can be easily and rapidly installed on the instrument panel of a cockpit of an aircraft (and also be easily and rapidly deinstalled), without modification of the instrument panel and equipment in the cockpit.
[0012] Said fastening system also has other advantages specified below.
[0013] In a preferred embodiment, the movable carriage is configured to be able to be brought, by

translational movement on the rail, into an end position on said rail, and the support device is able to pivot with respect to a panel of the movable carriage, about an axis which is transverse to said longitudinal axis, and it is configured to, by the pivoting thereof, be able, at least in said end position of the movable carriage, to be raised into what is referred to as a rest configuration.
[0014] Advantageously, said support device additionally comprises an auxiliary bracket, and the support bracket is able to pivot with respect to said auxiliary bracket by way of a hinge arranged at a first side, and it is configured to be able to be brought, by pivoting, into one or the other of the two following configurations: [0015] the lowered configuration, in which the support bracket is folded down on the auxiliary bracket so as to create between them a receptacle intended to receive the screen and the support bracket and the auxiliary bracket are secured to one another at a second side on the opposite side from said first side; and [0016] a raised configuration, in which the support bracket is raised with respect to the auxiliary bracket such that the support bracket and the auxiliary bracket are positioned substantially orthogonally with respect to one another.
[0017] In addition, advantageously, a first face of the auxiliary bracket and a first face of a plate connected to a panel of the movable carriage are in contact with one another and are connected to one another by way of, for the one part, a plurality of pins which are secured to at least one of said first faces and enter complementary and cooperating holes provided in the other of said first faces, and, for the other part, at least one double-sided adhesive strip, each face of which is adhesively bonded to one of said first faces of the auxiliary bracket and of the plate.
[0018] Furthermore, in a particular embodiment, the fastening system comprises a locking element configured to be able to lock the movable carriage in position on the rail.
[0019] Furthermore, advantageously, the fastening device comprises two fastening components arranged below the rail, each of said fastening components comprises a lower face intended to be fastened to one of the two footrests intended for a pilot of the aircraft and an upper face to which the rail is fastened, and is configured such that, when the fastening system is mounted on the instrument panel, the rail is positioned substantially horizontally.
[0020] Furthermore, in a particular embodiment, the fastening system comprises at least one bearing component arranged below the rail.
[0021] Advantageously, the fastening system also comprises a pivoting arm able to pivot about an axis transverse to said axis so as to be able to be brought into one or the other of the two following positions: [0022] a retracted position, in which it is retracted in a panel of the movable carriage; and [0023] an inclined position, in which a free end of the pivoting arm enters an opening in a plate of the support device in order to hold the support device in a rest configuration.
[0024] Furthermore, in a particular embodiment, the fastening system also comprises self-gripping attachment means for attaching together different elements of the fastening system.
[0025] The present invention also relates to a visualization system for an instrument panel of a cockpit of an aircraft, said visualization system comprising at least one screen.
[0026] According to the invention, said visualization system additionally comprises a system for removable fastening as described above, to which the screen is fastened and which is intended to be fastened to the instrument panel of the cockpit of the aircraft.

## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0027] The appended figures will make it easy to understand how the invention may be implemented. In these figures, identical reference signs denote similar elements.
[0028] FIG. **1** is a partial perspective view of an internal part of a cockpit of an aircraft, provided with a visualization system according to one embodiment of the invention, comprising a support device in a raised configuration, in an operation position.

[0029] FIG. **2** is a view similar to that of FIG. **1**, with the support device in a lowered configuration.
[0030] FIG. **3** is a side view of the visualization system with the support device in the raised configuration as in FIG. **1**.
[0031] FIG. **4** is a side view of the visualization system with the support device in the lowered configuration as in FIG. **2**.
[0032] FIG. **5** is a partial perspective view of an internal part of the cockpit of the aircraft, provided with the visualization system, with the support device, in a rest configuration, in an end position.
[0033] FIG. **6** is an exploded perspective view, as seen from below, of part of a rail, provided with a support device bearing a screen.
[0034] FIG. **7** is a view similar to that of FIG. **6**, as seen from slightly above.
[0035] FIG. **8** is a perspective view of a rail provided with a movable carriage and with a plate.
[0036] FIG. **9** is a transverse view of part of a rail, provided with a panel and with a locking element.
[0037] FIG. **10** is a perspective view of part of a rail, fastened to a footrest.
[0038] FIG. **11** is a slightly perspective view of a hinge of the support device, connecting a support bracket to an auxiliary bracket.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0039] The system **1** illustrating the invention and shown schematically, in a particular embodiment, in FIGS. **2** to **5** is intended to be mounted on an instrument panel **2** of a cockpit **3** of an aircraft, in particular of a transport plane.
[0040] More specifically, the system **1** is intended to fasten at least one screen **4** to the instrument panel **2**. The system **1** and the screen **4** form part of a visualization system **5**. The screen **4** can be any type of additional screen that a pilot of the aircraft may need, and notably a screen intended to be used when carrying out tests.
[0041] FIGS. **1** and **2** show part of the cockpit **3**, in the region of the seat of a pilot. The shown part of the cockpit **3** notably comprises the instrument panel **2**, two customary screens **6** and **7** (of which only the screen **7** is visible in FIG. **1**) and two footrest systems **10**A and **10**B. Each footrest system **10**A, **10**B comprises a footrest **9**A, **9**B, and a support **8**A, **8**B for supporting the footrest **9**A, **9**B, which is fastened in the cockpit **3**.
[0042] The system **1**, which can be arranged in the region of the seat of one or the other of the two pilots of the aircraft, comprises, as shown in FIG. **1**: [0043] a rectilinear rail **11** having a longitudinal axis A-A (FIGS. **1** and **8**); [0044] a fastening device **12** intended to (removably) fasten the rail **11** to the instrument panel **2**, without modification of said instrument panel **2**, as specified below; and [0045] at least one movable carriage **13** provided with a support device **14** intended to receive the screen **4**.
[0046] The description below relates to a system **1** provided with a single movable carriage **13** (comprising a support device **14** bearing a screen **4**). However, it is also conceivable, in the context of the present invention, for the system **1** to comprise a plurality of movable carriages (and notably two movable carriages) mounted on the same rail, each of the movable carriages comprising a support device bearing a screen.
[0047] In the context of the present invention: [0048] the adjectives “upper” and “lower” are defined with respect to a vertical direction, illustrated by an arrow Z in FIG. **2**, in the cockpit **3** when the system **1** is mounted on the aircraft, the adjective “upper” being defined in the direction of the arrow Z and the adjective “lower” being defined in the direction opposite to that of the arrow Z; and [0049] the adjective “horizontal” corresponds to a surface which is positioned substantially (+/−10%) horizontally when the aircraft is on the ground, namely which is orthogonal to the direction illustrated by the arrow Z.
[0050] The movable carriage **13** comprises a panel **15** corresponding to a longitudinal rectangular plate, which is able to slide on the rail **11**, and to which the support device **14** is connected, as

shown notably in FIGS. **3**, **4** and **6**.
[0051] The movable carriage **13** is configured to be able to be moved in translation along the rail **11** (in a direction parallel to the longitudinal axis A-A), as illustrated by a double-headed arrow F in FIGS. **1** and **8**, and it can be brought along said rail **11**, notably: [0052] into what is referred to as an operation position P**1**, as shown in FIGS. **1** and **2**; and [0053] into an end position P**2**, as shown in FIG. **5**.
[0054] The support device **14** comprises a support bracket **16** (FIGS. **3**, **4** and **7**) corresponding to a planar plate, which is intended to bear the screen **4**. To this end, the screen **4** is fastened in a customary manner to a face **16**A (FIGS. **3** and **4**) of the support bracket **16**. Preferably, the screen **4** is fastened by screws **17**, the heads of which are shown schematically in FIG. **7**.
[0055] In a preferred embodiment, the support bracket **16** (for example the position of holes for the passage of screws **17**) is adapted to fasteners provided on the screen **4** which is used. This makes it possible to easily adapt the system **1** to screens having different positions for the fastening points.
[0056] The support device **14** additionally comprises an auxiliary bracket **18** corresponding to a planar plate, as shown in FIGS. **3** and **4**.
[0057] The support bracket **16** is able to pivot with respect to the auxiliary bracket **18** by way of a customary hinge **19**, arranged at a side **20**A (FIG. **4**). The support bracket **16** is able to pivot, about an axis B-B, as illustrated by a double-headed arrow EB in FIGS. **1** and **7**. This axis B-B is parallel to said longitudinal axis A-A.
[0058] The support bracket **16** is configured to be able to be brought, by pivoting, into one or the other of the two following configurations: a lowered configuration and a raised configuration (or presentation configuration).
[0059] In the lowered configuration C**1**, shown in FIG. **4** in particular, the support bracket **16** is folded down on the auxiliary bracket **18** so as to create between them a receptacle **21** intended to receive the screen **4**.
[0060] In the lowered configuration C**1**, the support bracket **16** is connected to the auxiliary bracket at a side **20**B on the opposite side from said side **20**A such that they form a (partially open) support shell **22**.
[0061] As shown in FIGS. **6** and **7**, the support bracket **16** is provided, on the side **20**B, with a planar plate **23** connected by a bend to the support bracket **16**. A clamp **24** is fastened to the free end **23**A of this plate **23**.
[0062] The auxiliary bracket **18** is provided, on the side **20**B (on the opposite side from the side **20**A), with a projecting tab **25** provided with an opening **26**. The clamp **24** is configured to be able to be hooked in the opening **26**, as shown in FIG. **7**. To this end, the clamp **24** comprises two flexible arms **27**A and **27**B each provided, at the free end thereof, with a hook **28**A and **28**B. In the lowered configuration C**1**, the clamp **24** enters the opening **26** and is hooked, by way of its hooks **28**A and **28**B, at the edges of this opening **26**.
[0063] An (L-shaped) angle bracket **29**A, **29**B is connected to each flexible arm **27**A, **27**B. Pressure on the angle brackets **29**A and **29**B, in the direction illustrated by arrows I in FIGS. **6** and **7**, moves the arms **27**A and **27**B towards one another and releases the hooks **28**A and **28**B which can then be removed from the opening **26**. This makes it possible to raise the support bracket **16** (which pivots about the hinge **19**) and to disconnect the two brackets **16** and **18** on the side **20**B.
[0064] In the raised configuration C**2**, shown in FIG. **3**, the support bracket **16** is raised with respect to the auxiliary bracket **18** such that the support bracket **16** and the auxiliary bracket **18** are positioned substantially orthogonally with respect to one another.
[0065] Since the auxiliary bracket **18** remains arranged in a horizontal position on the rail **11** (as specified below), the support bracket **16** is thus brought into a vertical position. In this vertical position, a face **4**A of the screen **4** (fastened to the face **16**A of the support bracket **16**) is visible, as shown in FIGS. **1** and **3**. The face **4**A is the face for displaying or presenting information of the screen **4**. In this raised configuration C**2**, the screen **4** can therefore be used by the pilot.

[0066] In order to pass from the lowered configuration C**1** to the raised configuration C**2**, the support bracket **16** is therefore pivoted with respect to the auxiliary bracket **18** by way of the pivot hinge **19**, provided with two articulations **58**A and **58**B. As shown in FIG. **11**, the hinge **19** comprises tabs **59**A and **59**B which are connected to the support bracket **16** and cooperate, respectively, with tabs **60**A and **60**B connected to the auxiliary bracket **18**. At the articulation **58**A, a washer **61** is arranged between the tabs **59**A and **60**A, and at the articulation **58**B, a washer **62** is arranged between the tabs **59**B and **60**B. A connecting component **63** which passes through the holes in the tabs **59**B and **60**B and in the washer **62** connects together these components. At the articulation **58**A, the tabs **59**A and **60**A and the washer **61** are connected together by way of a screw adjuster **64** provided with a nut **65**.
[0067] The screw adjuster **64**, by being screwed, generates a stress on the articulation **58**A in order to lock its movement and thus prevent the pivoting of the support bracket **16** with respect to the auxiliary bracket **18**. The screw adjuster **64** is configured to prevent that the auxiliary bracket **18** pivots with respect to the support bracket **16** in the raised configuration C**2** and remains in its vertical position.
[0068] In a particular embodiment, the washers **61** and **62** comprise thrust balls for generating a position-maintaining force in a predefined relative position between the two brackets **16** and **18**, in the present case when they are positioned substantially orthogonally with respect to one another in the raised configuration C**2**.
[0069] Furthermore, the support device **14** additionally comprises a plate **30**, as shown in FIGS. **6** and **7** in particular.
[0070] The plate **30** is connected by a lower face **30**A (FIG. **6**) to an upper face **15**B (FIG. **7**) of the panel **15**, as specified below.
[0071] In addition, the auxiliary bracket **18** of the support shell **22** is connected to said plate **30**. More specifically, the lower face **18**A (FIG. **6**) of the auxiliary bracket **18** and an upper face **30**B (FIG. **7**) of the plate **30** are in contact with one another. These faces **18**A and **30**B are connected to one another by way of a plurality of pins **31** (FIG. **7**) and at least one double-sided adhesive strip **32** (FIG. **6**), that is to say a strip whose two faces are adhesive.
[0072] One of the adhesive faces of the double-sided adhesive strip **32** is thus adhesively bonded to the lower face **18**A of the auxiliary bracket **18** and the other adhesive face of the double-sided adhesive strip **32** is adhesively bonded to the upper face **30**B of the plate **30**. In the example of FIG. **6**, two adhesive strips **32**, of rectangular shape, are provided on the face **18**A of the auxiliary bracket **18**.
[0073] Furthermore, as shown in FIG. **7**, the pins **31** are fastened to the upper face **30**B of the plate **30** and, in the assembled position of the auxiliary bracket **18** and of the plate **30**, the pins **31** enter complementary and cooperating holes **33** (FIG. **6**) provided in the lower face **18**A of the auxiliary bracket **18**. Of course, it is also conceivable to provide the pins on the face **30**B and cooperating holes on the face **18**A, or to provide pins on the two faces **18**A and **30**B, which are associated with holes provided in each case on the opposite face.
[0074] The means for attaching the support shell **22** (provided with the screen **4**) to the plate **30** connected to the rail **11**, which comprise the pins **31** and the adhesive strips **32**, make it possible:
[0075] for the one part, to provide an attachment sufficient for the different operations to be carried out; and [0076] for the other part, to detach or tear off the support shell **22** (and the screen **4**) from the system **1**, by manually pulling the support shell **22** in the direction of the arrow Z in the lowered configuration (FIG. **2**).
[0077] This makes it possible, notably in the event of an emergency, for the pilot to very rapidly remove the screen **4** (surrounded by the support shell **22**) from the instrument panel **2**.
[0078] In a preferred embodiment, notably the auxiliary bracket **18** and the plate **30** are produced by an additive manufacturing process, making it possible to reduce the manufacturing cost of these components and thus of the system **1**. In addition, these components can, in this case, be produced

easily.

[0079] As indicated above, the movable carriage **13** is configured to be able to be moved in translation along the rail **11**, as illustrated by a double-headed arrow F in FIGS. **1** and **8**, and it can be brought along said rail **11**, notably: [0080] into the operation position P**1** shown in FIGS. **1** and **2**; and [0081] into the end position P**2** shown in FIG. **5**.

[0082] The movable carriage **13** can thus be brought into the desired position by the pilot, in order for the screen **4** to be disposed exactly at the location desired by the pilot in the raised configuration, i.e., preferably within their field of vision when they are seated on their seat in order to prevent them having to turn their head when they wish to view the screen **4**.

[0083] To this end, in a particular embodiment, the rail **11**, which is for example produced from metal and notably from aluminum, comprises, as shown in FIG. **8**, three rectilinear guides **34**A, **34**B and **34**C which are parallel to the longitudinal axis A-A, namely a central guide **34**C and two guides **34**A and **34**B arranged on either side of this central guide **34**C.

[0084] The three guides **34**A, **34**B and **34**C (FIG. **9**) have a similar shape in cross section, comprising on the upper face **11**B of the rail **11** a longitudinal slot **35**, leading into a longitudinal groove **36** of substantially rectangular general shape.

[0085] The panel **15**, for its part, is provided on its lower face **15**A with two longitudinal ribs **37**A and **37**B. These longitudinal ribs **37**A and **37**B have a rectangular general shape in cross section, which is complementary to that of the grooves **36**, and they are inserted in the grooves **36** of the two guides **34**A and **34**B.

[0086] This cooperation of shape between the ribs **37**A and **37**B and the grooves **36** allows the panel **15** to be held on the rail **11** and to be able to be moved longitudinally on the rail **11**.

[0087] Furthermore, in a particular embodiment, the system **1** also comprises a locking element **38** (FIG. **9**) for locking the panel **15** and therefore the movable carriage **13** in position on the rail **11**, that is to say for preventing the movement of the movable carriage **13** when it is locked.

[0088] The locking element **38** comprises a screw **39** provided with a head **39**A (comprising a gripping plate **39**B) and with a threaded stem **39**C, as shown in FIG. **9**.

[0089] The head **39**A of the locking element **38** rests on the upper face **15**B of the panel **15** and the threaded stem **39**C passes through the slot **35** of the guide **34**C and ends in the groove **36** of the guide **34**C. A nut **40** is housed in this groove **36** and is screwed on the threaded stem **39**C.

[0090] Thus: [0091] in order to lock the movable carriage **13** in position on the rail **11**, it suffices for the locking element **38** to be screwed (by manually turning the gripping plate **39**B) until clamping between the nut **40** and the head **39**A is obtained that is sufficient to hold the panel **15** against the rail **11**; and [0092] in order to unlock the movable carriage **13** so as to allow it to move in translation along the rail **11**, it suffices to loosen the locking element **38** enough to allow the panel **15** to be able to be moved with respect to the rail **11**.

[0093] Furthermore, the rail **11** also comprises, as shown in FIGS. **8** and **9**, two rectilinear guides **34**D and **34**E which are parallel to the longitudinal axis A-A and are intended for the fastening of the rail **11** as specified below.

[0094] In the embodiment shown, the guides **34**D and **34**E have a similar shape in cross section to that of the guides **34**A, **34**B and **34**C, comprising on the lower face **11**A of the rail **11** a longitudinal slot **35**, leading into a groove **36** of substantially rectangular general shape.

[0095] Furthermore, the fastening device **12** comprises two fastening components **41** arranged below the rail **11**, in the region respectively of the supports **8**A and **8**B of the footrests **9**A and **9**B.

[0096] The function of these fastening components **41** is to fasten the system **1** (bearing the screen **4**) to the aircraft. One of the fastening components **41** is intended to be fastened to the support **8**A of the footrest **9**A and the other fastening component **41** is intended to be fastened to the support **8**B of the footrest **9**B.

[0097] As shown in FIG. **10**, each fastening component **41** comprises a central plate **42** and two lateral plates **43** and **44** arranged on either side of the central plate **42** and connected to the latter, respectively, by tabs **45** and **46**.

[0098] Each of the lateral plates **43** and **44** is provided with a hole **43**A, **44**A and is able to be fastened by way of a bolt **47** to the support **8**A, **8**B of a footrest **9**A, **9**B.

[0099] Customarily, each support **8**A, **8**B of a footrest **9**A, **9**B comprises two pre-existing holes, which are intended to receive a cover which is removed before fastening the fastening component **41**. The holes **43**A and **44**A in the lateral plates **43** and **44** of the fastening component **41** are formed and positioned so as to be able to be superposed with the pre-existing holes in the supports **8**A and **8**B.

[0100] Thus, the rail **11** and therefore the system **1** can be fastened to the aircraft without needing to modify the instrument panel **2** or equipment in the cockpit of the aircraft, by using holes that already exist in the supports **8**A and **8**B of the footrests **9**A and **9**B.

[0101] In addition, two holes **42**A and **42**B are provided in the central plate **42** in order to allow it to be fastened below the rail **11** with the aid of two fastening elements **48**, preferably bolts, which pass through these holes **42**A and **42**B and comprise nuts **66** arranged respectively in the grooves **36** of the guides **34**D and **34**E, as shown in FIG. **9**.

[0102] The upper face **56**A, **56**B of each support **8**A, **8**B of a footrest **9**A, **9**B does not have a horizontal surface. Thus, the fastening components **41** arranged on these upper faces **56**A and **56**B are formed so as to position the rail **11** horizontally.

[0103] To this end, for each of the fastening components **41**, the upper face **41**B (of the central plate **42**) and the lower faces **41**A (of the lateral plates **43** and **44**), illustrated in FIG. **4** respectively by dashed line segments D**1** and D**2**, form between them a non-zero angle α. This angle α is such that, when the system **1** is installed on the instrument panel **2**, the rail **11** is in a horizontal position.

[0104] Consequently, the fastening components **41** are configured so as: [0105] to be able to be fastened to the supports **8**A and **8**B of the footrests **9**A and **9**B; and [0106] to be able to be fastened below the rail **11**, in order to position it horizontally.

[0107] Furthermore, the system **1** also comprises a bearing component **49** shown in FIGS. **1** and **2**. This bearing component **49** is arranged between the lower face of the rail **11** and the upper face of a customary bracket **50** of the instrument panel **2**, which is in a folded position, in order to provide additional holding of the rail **11**.

[0108] Furthermore, the plate **30** is connected by way of a customary hinge **51** to the panel **15**, as shown in FIG. **5**.

[0109] As indicated above, the movable carriage **13** is configured to be able to be brought, by translational movement on the rail **11**, into an end position P**2** on said rail **11**, such as the position shown in FIG. **5**.

[0110] In the end position P**2**, the plate **30** of the support device **14** is able to pivot with respect to the panel **15** of the movable carriage **13**, about an axis C-C (which is transverse, and preferably orthogonal, to said longitudinal axis A-A), as illustrated by a double-headed arrow EC in FIG. **5**.

[0111] In this end position P**2**, the support device **14** is thus configured to be able to be brought, by the pivoting of the plate **30** about the axis C-C, into one or the other of the two following configurations: [0112] the lowered configuration C**1**, in which the plate **30** is in contact with the panel **15** and the support device **14** is therefore lowered on the movable carriage **13**; and [0113] a rest configuration C**3**, in which the plate **30** and therefore the support device **14** are raised, as shown in FIG. **5**.

[0114] In this rest configuration C**3**, the support device **14** and the screen **4** are positioned so as to hinder the pilot as little as possible, if at all.

[0115] The support device **14** is held in the lowered configuration C**1**, with the aid of self-gripping attachment means **52** (FIG. **5**). The "scratch" self-gripping attachment means **52** are provided between the plate **30** and the panel **15** and have the function of attaching the plate **30** to the panel **15**.

[0116] Furthermore, in the rest configuration C**3**, the support device **14** is held in position with the

aid of a pivoting arm **53** (FIGS. **5** and **8**) and self-gripping attachment means **54** (FIG. **6**).
[0117] The “scratch” self-gripping attachment means **54** are provided between the auxiliary bracket **18** and a lateral edge of the rail **11**.
[0118] The pivoting arm **53**, preferably produced in the form of a rectangular planar plate, is able to pivot about an axis D-D (which is transverse, and preferably orthogonal, to said longitudinal axis A-A), as illustrated by a double-headed arrow ED in FIG. **8**.
[0119] The pivoting arm **53**, which is connected at one end **53**A by way of a customary hinge to the panel **15**, can be brought into one or the other of the two following positions: [0120] a retracted position, in which it is retracted in a receptacle **55** (FIGS. **5** and **8**) corresponding to a cutout provided in the panel **15**; and [0121] an inclined (support) position as shown in FIGS. **5** and **8**, in which a free end **53**B of the pivoting arm **53** enters an opening **57** (FIG. **7**) in the plate **30** of the support device **14** in order to hold the support device **14** in said rest configuration C**3**.
[0122] Methods for installing and deinstalling the system **1** will now be described.
[0123] The method for installing the system **1** on the instrument panel **2** of an aircraft, so as to be usable by a pilot, notably comprises the following (manual) operations: [0124] disposing the rail **11** in the appropriate position on the instrument panel **2**; [0125] installing the bearing component **49** between the rail **11** and the bracket **50**; [0126] fastening the two fastening components **41** of the fastening system **12** to the supports **8**A and **8**B of the two footrests **9**A and **9**B intended for the pilot, by screwing the bolts **47** in the pre-existing holes in the supports **8**A and **8**B. Preferably, for this last operation, the fastening components **41** (which are screwed to the rail **11** beforehand) are simply screwed to the supports **8**A and **8**B. In a variant, it is also conceivable for the fastening components **41** (which are screwed to the supports **8**A and **8**B beforehand) to be screwed to the rail **11** during this last operation.
[0127] Furthermore, the method for deinstalling the system **1** (previously installed on the instrument panel **2**) notably comprises the following (manual) operations: [0128] unscrewing the bolts **47** in order to separate the two fastening components **41** of the system **1** from the supports **8**A and **8**B of the footrests **9**A and **9**B to which they were fastened; [0129] removing the rail **11** (and all the components that it bears); [0130] removing the bearing component **49**.
[0131] Consequently, the installation and the deinstallation of the system **1** can be carried out easily and rapidly, with a reduced number of operations, and additionally without needing to modify the instrument panel **2** or equipment in the cockpit **3** of the aircraft.
[0132] Furthermore, in order to move the movable carriage **13** (provided with the support device **14** and the screen **4**) from a first position to a second position, for example from the operation position P**1** to the end position P**2** or vice versa, it suffices to carry out the following (manual) operations: [0133] in the first position, loosening the locking element **38** in order to release the panel **15** from the rail **11**; [0134] moving the movable carriage **13** by causing it to slide on the rail **11** in order to bring it into the second position; and [0135] in this second position, screwing the locking element **38** in order to prevent any movement of the movable carriage **13**.
[0136] By virtue of the rail **11** which is arranged on the instrument panel **2**, the movable carriage **13** can be brought to any desired position along said rail **11**, and in particular to the end position P**2** and to the operation position P**1** (in which the screen **4** is positioned in front of the customary screen **6** of the cockpit **3** in the example of FIGS. **1** and **2**).
[0137] The system **1** thus provides a great degree of lateral positioning flexibility for the screen **4** in the cockpit **3**.
[0138] Methods for bringing the support device **14** of the system **1** into different possible configurations will now be described.
[0139] The method for, notably in the operation position Pl on the rail **11**, causing the support device **14** to pass from the raised configuration C**2** in FIG. **1** to the lowered configuration C**1** in FIG. **2** notably comprises the following (manual) operations: [0140] acting on the screw adjuster **64** in order to allow the support bracket **16** to pivot via the hinge **19**; [0141] moving the support

bracket **16** bearing the screen **4** by causing it to pivot about the axis B-B in order to bring it from the vertical position (relative to the raised configuration C**2**) to a horizontal position (in which it is in contact with the auxiliary bracket **18**), the auxiliary bracket **18** resting on the plate **30** which itself rests on the panel **15** connected to the rail **11**; and [0142] introducing the clamp **24** into the opening **26** in order to secure the support bracket **16** to the auxiliary bracket **18**.

[0143] Furthermore, the method for, notably in the operation position Pl on the rail **11**, causing the support device **14** to pass from the lowered configuration C**1** in FIG. **2** to the raised configuration C**2** in FIG. **1** notably comprises the following (manual) operations: [0144] removing the clamp **24** from the opening **26** in order to disconnect the support bracket **16** from the auxiliary bracket **18**; [0145] moving the support bracket **16** bearing the screen **4** by causing it to pivot about the axis B-B in order to bring it from the horizontal position (in which it is in contact with the auxiliary bracket **18**) to the vertical position (relative to the raised configuration C**2**); and [0146] acting on the screw adjuster **64** in order to prevent pivoting of the support bracket **16**.

[0147] Consequently, the change of configuration of the support device **14**, notably in the operation position P**1**, can be carried out easily and rapidly, with a reduced number of operations.

[0148] Furthermore, the method for, in the end position P**2** on the rail **11**, causing the support device **14** to pass from a lowered configuration C**1** such as the one in FIG. **2** to the rest configuration C**3** in FIG. **5** notably comprises the following (manual) operations: [0149] moving the support device **14** bearing the screen **4** by causing it to pivot about the axis C-C in order to bring it from the horizontal position (in which it rests on the rail **11**) to the vertical position (relative to the rest configuration C**3**); and [0150] causing the pivoting arm **53** to pivot about the axis D-D in order to bring it from the retracted position to the inclined position in which its free end **53**B enters the opening **57** in the plate **30**. In this inclined position, the pivoting arm **53** holds the support device **14** in the (vertical) rest configuration.

[0151] Furthermore, the method for, in the end position P**2** on the rail **11**, causing the support device **14** to pass from the rest configuration C**3** in FIG. **5** to a lowered configuration C**1** such as the one in FIG. **2** notably comprises the following (manual) operations: [0152] causing the pivoting arm **53** to pivot from the inclined position to the retracted position so as to permit the pivoting of the support device **14**; and [0153] moving the support device **14** bearing the screen **4** by causing it to pivot about the axis C-C in order to bring it into the horizontal position (relative to the lowered configuration C**1**).

[0154] Consequently, the change of configuration of the support device **14**, in the rest position P**2**, can be carried out easily and rapidly, with a reduced number of operations.

[0155] The system **1** as described above has a very large number of advantages. In particular: [0156] it can be installed on the instrument panel, without modification of the instrument panel and equipment in the cockpit of the aircraft; [0157] the installation and the deinstallation of the system **1** can be carried out easily and rapidly; [0158] the operations for installation and deinstallation are operations that are simple and few; [0159] the system **1** has a reduced cost; [0160] the support device and the screen can be brought into the desired position along the rail, and in particular into a position adapted to the field of vision of the pilot; [0161] when the pilot no longer needs the screen, the support shell and the screen that it bears can either be brought into the rest configuration or be removed rapidly in the event of an emergency, such that they do not hinder the pilot; [0162] the system **1** can be used for a screen intended for the pilot or for a screen intended for the co-pilot; and [0163] the system **1** can be used for any type of additional screen, and in particular for a test screen.

[0164] While at least one exemplary embodiment of the present invention(s) is disclosed herein, it should be understood that modifications, substitutions and alternatives may be apparent to one of ordinary skill in the art and can be made without departing from the scope of this disclosure. This disclosure is intended to cover any adaptations or variations of the exemplary embodiment(s). In addition, in this disclosure, the terms “comprise” or “comprising” do not exclude other elements or steps, the terms “a” or “one” do not exclude a plural number, and the term “or” means either or both. Furthermore, characteristics or steps which have been described may also be used in combination with other characteristics or steps and in any order unless the disclosure or context suggests otherwise. This disclosure hereby incorporates by reference the complete disclosure of any patent or application from which it claims benefit or priority.

## Claims

**1**. A system for removably fastening at least one screen to an instrument panel of a cockpit of an aircraft, the system comprising: a rectilinear rail having a longitudinal axis; a fastening device configured to fasten the rectilinear rail to an instrument panel without modification of said instrument panel; and at least one movable carriage provided with a support device, said at least one movable carriage configured to move in translation along said rectilinear rail and to be brought into an operation position, said support device being provided with a support bracket configured to bear a screen, the support bracket configured to pivot about an axis parallel to said longitudinal axis and being configured to, by the pivoting thereof, be raised into a raised configuration in said operation position, and to be able to be lowered on the movable carriage into a lowered configuration, at least during a translational movement of said movable carriage on the rail.
**2**. The system according to claim 1, wherein the movable carriage is configured to be brought, by translational movement on the rectilinear rail, into an end position on said rectilinear rail, and wherein the support device is configured to pivot with respect to a panel of the at least one movable carriage, about an axis which is transverse to said longitudinal axis, and, by the pivoting thereof, is configured to, in said end position, be raised into a rest configuration.
**3**. The system according to claim 1, wherein the support device further comprises an auxiliary bracket, and wherein the support bracket is configured to pivot with respect to said auxiliary bracket by a hinge arranged at a first side, and further configured to, by pivoting, be brought into one or the other of the two following configurations: the lowered configuration, in which the support bracket is folded down on the auxiliary bracket so as to create a receptacle configured to receive the screen and the support bracket and the auxiliary bracket are secured to one another at a second side on the opposite side from said first side; and a raised configuration, in which the support bracket is raised with respect to the auxiliary bracket such that the support bracket and the auxiliary bracket are positioned substantially orthogonally with respect to one another.
**4**. The system according to claim 3, wherein a first face of the auxiliary bracket and a first face of a plate connected to a panel of the movable carriage are in contact with one another and are connected to one another by a plurality of pins which are secured to at least one of said first faces and enter complementary and cooperating holes provided in the other of said first faces, and, for the other of said first faces, at least one double-sided adhesive strip, each face of which is adhesively bonded to one of said first faces of the auxiliary bracket and of the plate.
**5**. The system according to claim 1, further comprising: a locking element configured to lock the at least one movable carriage in position on the rectilinear rail.
**6**. The system according to claim 1, wherein the fastening device comprises two fastening components arranged below the rectilinear rail, wherein each of said fastening components comprises a lower face configured to be fastened a footrests of the aircraft and an upper face to which the rectilinear rail is fastened, and wherein the two fastening components are configured such that, when the fastening system is mounted on the instrument panel, the rectilinear rail is positioned substantially horizontally.
**7**. The system according to claim 1, further comprising: at least one bearing component arranged below the rectilinear rail.
**8**. The system according to claim 1, further comprising: at least one pivoting arm configured to pivot about an axis transverse to said longitudinal axis so to be brought into one or the other of the two following positions: a retracted position, in which the at least one pivoting arm is retracted in a

panel of the movable carriage; and an inclined position, in which a free end of the at least one pivoting arm enters an opening in a plate of the support device in order to hold the support device in a rest configuration.

**9**. The system according to claim 1, further comprising: self-gripping attachment means for attaching together different elements of the system.

**10**. A removable visualization system for an instrument panel of a cockpit of an aircraft, said visualization system comprising: at least one screen, and, the system according to claim 1, wherein the at least one screen is fastened to an instrument panel of a cockpit of an aircraft.